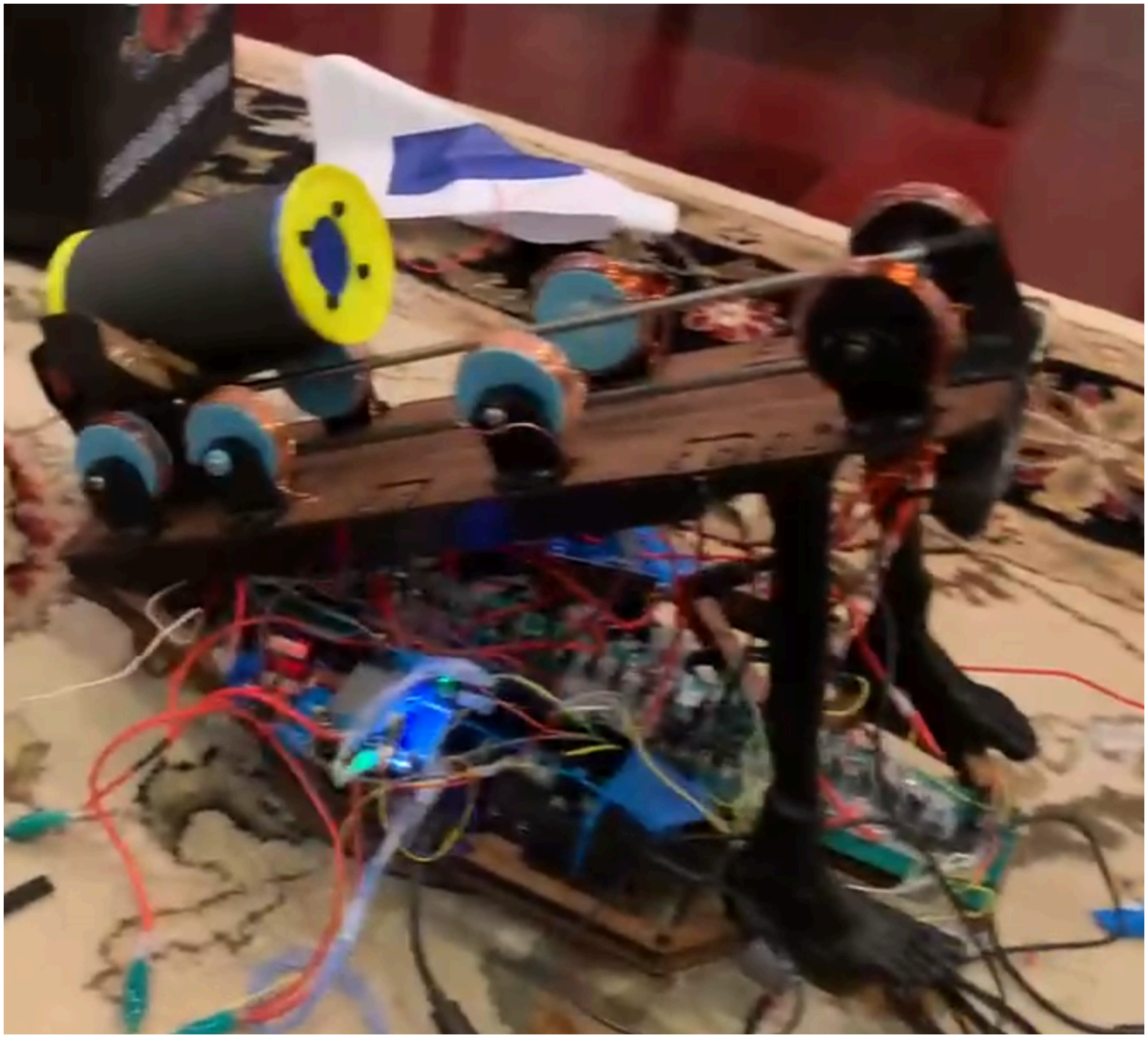
Group 5: Coil-Powered Trash Sorter



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Project Overview

Criteria 1, 3, 11

Purpose

This is an expansion upon the previous semester's project. The objective is to dispose of waste in the correct categories.
As we wrote in the last report, "just in the past year, Americans have only a 32% recycling rate, leaving more than 2/3 of recyclable materials lost. Furthermore, not only are potential resources wasted, but 25% of recycling bins are contaminated, leading to the entire batch being unrecyclable. It is estimated that over 8 million metric tons of plastic enter the oceans annually, which is the equivalent of dumping in a garbage truck full of plastic every minute."

Last semester's project focused on sorting white and yellow balls to represent trash and recycling, but to improve upon that concept, we wanted to create a prototype that wasn't limited to a specifically sized spherical object that had to be a certain mass, as we discovered it wasn't adaptable enough. This led us to develop a new capsule in the form of a cylindrical shape, and a launching mechanism that is versatile and could adapt to various capsule shapes depending on the carriage shape. The waste would be placed into the capsule and loaded onto the carriage, and sorted using the turret mechanism into the bins.

Existing Knowledge and Research

Criteria 2

Trash Sorting Robots

- One notable robot uses existing robotic arms and adding depth-sensing capable cameras to identify different types of trash. More on sorting algorithms will be mentioned under the “Sorting Algorithms” section. The cameras also are able to predict optimal positions on a piece of waste to grab it properly, without significantly making specialized grippers for types of trash. It shows that it is not necessarily mandatory to re-develop whole hardware systems just for the purpose of trash sorting (Bacchin et al. 2024).
- **Another system by AMP Robotics is already being implemented in the real world. With its fully-automatic trash sorting using AI detection with ~99% accuracy, it also uses robotic arms to pick up and relocate trash.** Their design is also modular and thus can adapt to various facility sizes and requirements.
- A commercial startup example is CleanRobotics' TrashBot. **However, what distinguishes this from our robot is that it sorts waste where it is disposed of**. One advantage of their approach is that they could provide real-time feedback to the general population. Once their robot sorts the item, it will give “a helpful tip based on the item they just threw away.” **This allows it to have a greater impact beyond the disposal site, as people using the robot would be encouraged to engage in better waste practices.**

Coil Launchers

- Our approach consists of using a coil launcher for repeatability and reliability, as moving parts tend to need more maintenance, which can be detrimental if the goal is to sort rapidly and frequently.
- One example of a coil launcher is by Einat and Orbach (2023). They designed a multi-stage coilgun, which is able to launch a 2.5-gram ferromagnetic projectile up to 130 m/s. There are key differences in our electromagnetic launchers, as we have to accommodate larger and heavier payloads. While it is advantageous to have the projectile at the center of the coil, where the magnetic field is more concentrated, it is not possible with our objective. Their research is generally aimed at the application toward lunar launches; however, it is applicable to systems requiring repeatable and fuel-free actuation.
- One thing we didn’t think of is the use of energy-recycling ignition circuit. This actually improves the launcher’s efficiency while also reducing the current pulse duration. The reason this is important is that we are both using reluctance systems;

the timing of each stage's pulse will change whether the projectile is repelled forward or attracted back. While we use a traditional flyback diode approach, they allows the capacitor to recharge in reverse polarity. A future step in experimentation would be this pulse control, as it could be directly adapted to our coil launcher with small modifications.

Sorting Algorithms

- Our project hinges upon a low-cost, real-time object detection system for sorting, which is currently based on colored markers; however, the camera used for sorting could also be used for a more versatile set of trash/recycling bins. It could also be used to classify the trash itself. An example of a real-time object detection system is by Dong et al. (2022), which explores the feasibility of deploying deep learning models for object detection on Raspberry Pi platforms. They employ SSD-MobileNet, which is a lightweight, single-shot detection algorithm. The mobile-optimized deep learning models could classify and find objects, people, and laptops while running in real time. While we opted for a simpler approach due to a time constraint, a next step could be leveraging public datasets on types of trash with this algorithm, allowing us to sort without human intervention.

- To improve the accuracy of such a model, Domínguez-Almaraz, Mucientes, and Bugarín (2021) have a solution with a two-stage deep learning framework. Their framework separates object detection from classification (identifying waste type), which can widen the possible data set. First, they use the **EfficientDet-D2** model for detection, as it's more accurate and computationally efficient. Then, using they use the **EfficientNet-B2** for classification. Their approach has better performance across different environments (indoor, outdoor, and underwater), which is suited for our project, as we plan to deploy a version of our robot in outdoor environments and not have it confined to a waste facility.

Investigations

INVESTIGATION 1: Capacitor Charging/Discharging

Concepts Covered:

- Capacitors in Series and Parallel
- Potential Energy of a Capacitor
- Capacitor and Energy Storage Circuits
- Electric Field Energy Density (in the supercap)
- Resistors in Series and Parallel
- Charging/discharging time constant of RC circuit
- ESR of a capacitor

Purpose

Determine the ESR_{cap}, R_{COIL}, and Capacitance using RC time constants.

Procedure

Charging with a buck boost converter does not demonstrate RC characteristics, we will focus on analyzing the discharge. Our R_total is equal to $ESR_{cap} + 2 * R_{COIL}$. In the time constant formula we have 3 unknowns, so we must record 3 sets of data to have 3 equations to solve for these

Part A: 2 Coils

1. Charge the capacitors to 15.6V
2. Connect the positive and negative ends of oscilloscope to either end of the capacitor
3. Record the voltage drop and time constant

Part B: 1 Coil

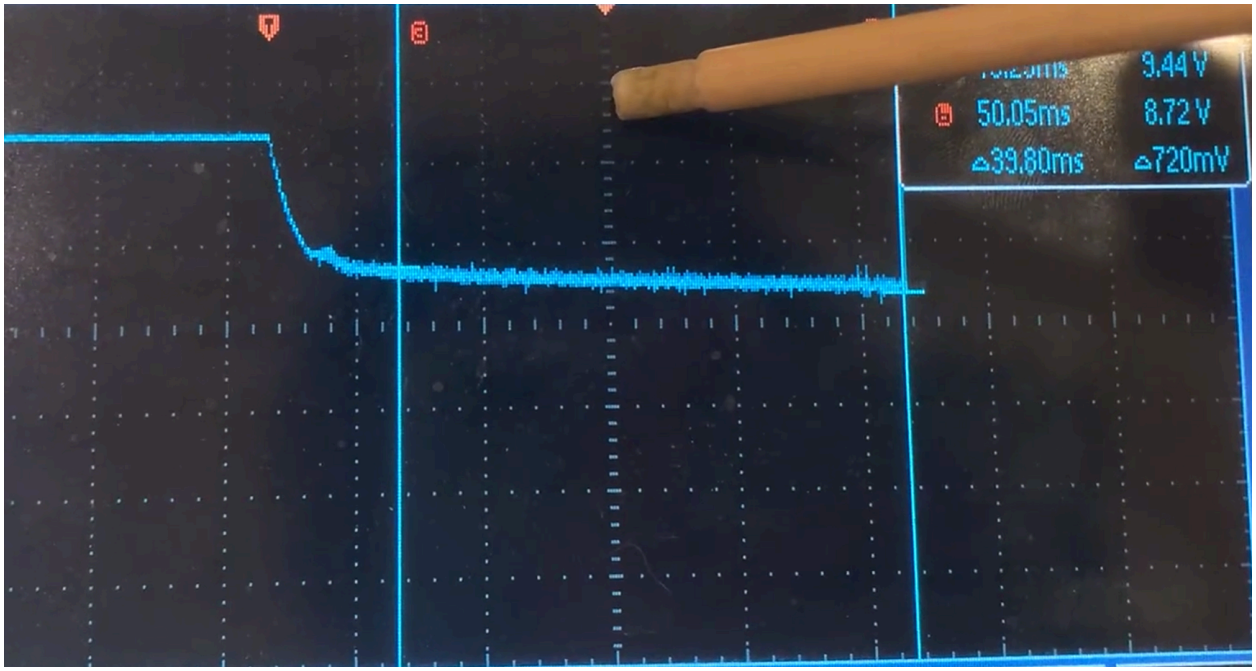
4. Cut one coil
5. Charge the capacitors to 15.6V
6. Connect the positive and negative ends of oscilloscope to either end of the capacitor
7. Record the voltage drop and time constant

Part C: 2 Coils + 1 Ohm

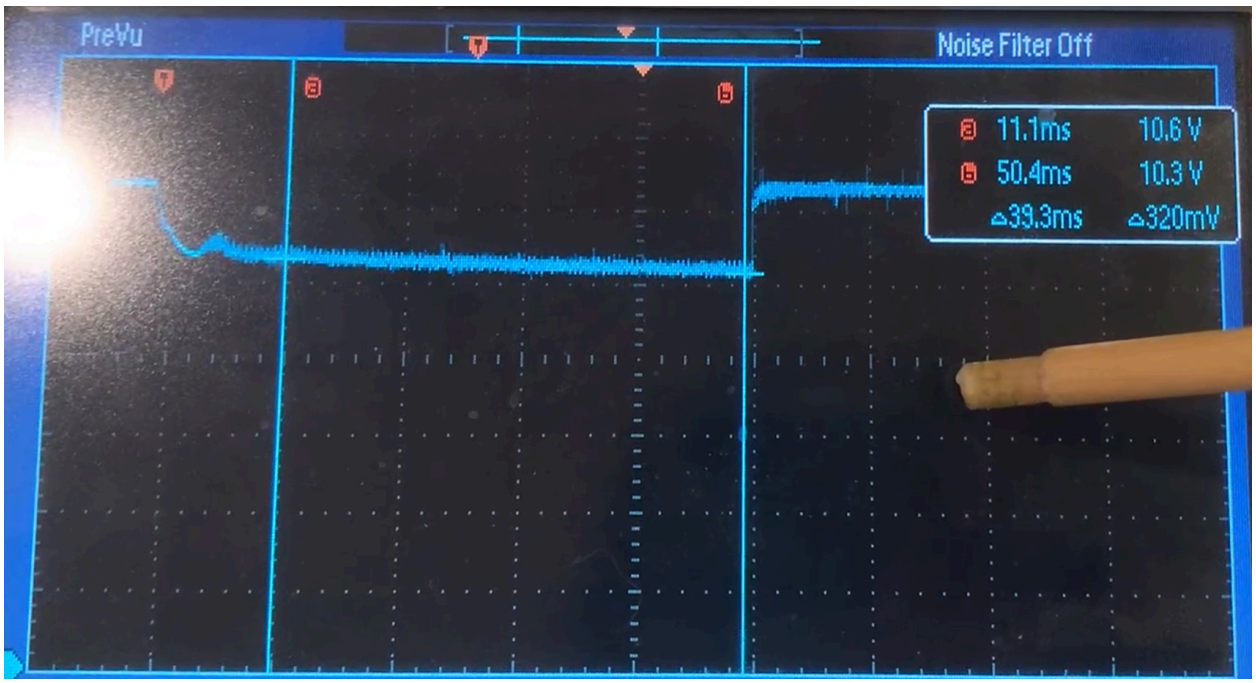
8. Attach a 1 Ohm resistor between the cut coil
9. Charge the capacitors to 15.6V
10. Connect the positive and negative ends of oscilloscope to either end of the capacitor
11. Record the voltage drop and time constant

Data

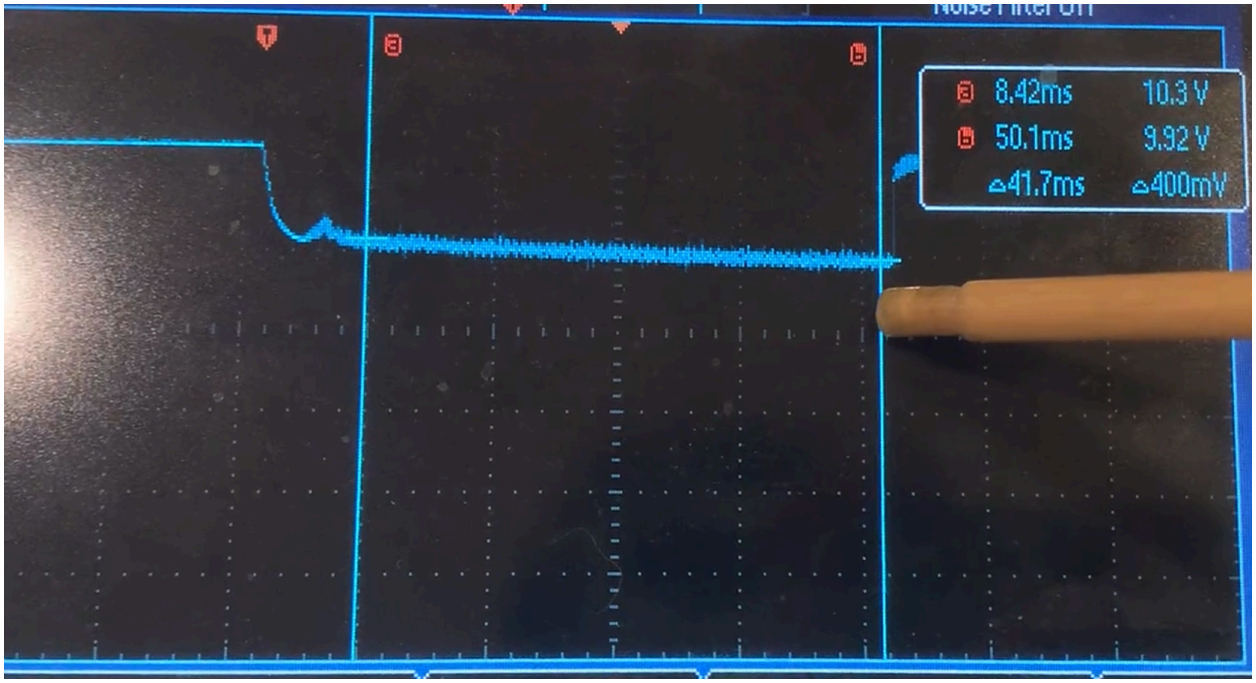
Two Coils RC Discharging V(t) Graph



One Coil RC Discharge V(t) Graph



Two Parallel Coils + 1 ohm series resistor RC Discharge V(t) Graph



Data Analysis

(ESR, Series/Parallel Resistors, RC Circuit Time Constants)

Step 1: Calculate Time Constants

Time constant τ is calculated by:
$\tau = - \Delta t / \ln(V_f / V_0)$
Δt = time interval, V_0 = initial voltage, V_f = final voltage

Case 1: $R = R_{ESR} + 2 * R_{COIL}$
$\tau_1 = - \Delta t_1 / \ln(V_{f_1} / V_{0_1})$
$\Delta t_1 = 0.0398$ s
$V_{0_1} = 9.44$ V
$V_{f_1} = 8.72$ V

Case 2: $R = R_{ESR} + R_{COIL}$
$\tau_2 = - \Delta t_2 / \ln(V_{f_2} / V_{0_2})$
$\Delta t_2 = 0.0393$ s
$V_{0_2} = 10.6$ V
$V_{f_2} = 10.3$ V

Case 3: $R = R_{ESR} + R_{COIL} + (R_{COIL} \,||\, 1\ \Omega)$

$\tau_3 = - \Delta t_3 / \ln(V_f_3 / V_0_3)$
$\Delta t_3 = 0.0417$ s
$V_0_3 = 10.3$ V
$V_f_3 = 9.92$ V

Calculations
$\tau_1 = -0.0398 / \ln(8.72 / 9.44) = 0.50$ s
$\tau_2 = -0.0393 / \ln(10.3 / 10.6) = 1.36$ s
$\tau_3 = -0.0417 / \ln(9.92 / 10.3) = 1.10$ s

Expected Time Constants

Expected time constants with R_ESR = 0.18 Ω, R_COIL = 0.32 Ω, C = 16.5 F:

Case 1: R = R_ESR + 1/2 * R_COIL
$\tau_1 = (0.18 + 1/2 * 0.32) * 16.5 = (0.18 + 0.16) * 16.6 = 0.34 * 16.6 = 5.64$ s

Case 2: R = R_ESR + R_COIL
$\tau_2 = (0.18 + 0.32) * 16.5 = 0.50 * 16.5 = 8.25$ s

Case 3: R = R_ESR + R_COIL || (R_COIL + 1 Ω)
First calculate R_COIL || 1 Ω:
R_parallel = (0.32 * 1.32) / (0.32 + 1.32) ≈ 0.258 Ω

Then total resistance
R = 0.18 + 0.26 = 0.44 Ω

$\tau_3 = 0.44 * 16.5 \approx 7.26$ s

Error Analysis

The expected vs observed time constant has a very high 85% error, indicating that something must have been wrong with our assumptions. We believe that the source of error would most likely be that the ESR is non constant and increases as we discharge and the capacitor and joints temperature increase. This is likely due to our high 30A current draw causing things to heat up and change. Additionally, any damage we caused to the capacitor in our weeks of use would have been seen here. We initially discharged the capacitor with a screwdriver, this high amperage may have caused some unintended damage as seen in the 4th cell in Investigation 2.

Capacitors in Series & Parallel

What Was Justified

The equivalent capacitance of a capacitor within a system is 1/Ceq = 1/C1 + 1/C2 + … + 1/Cn for series and Ceq = C1 + C2 + … Cn for parallel

Evidence

Computational

SERIES

Our capacitor bank consists of six 100F capacitors in series. Using the formula for equivalent capacitance, 1/Ceq = 1/C1 + 1/C2…1/C6 we get

- 1/Ceq = 6(1/100F)
- 1/Ceq = 0.06
- Ceq = 1/0.06 = 16.6F

Super Capacitor 6pcs/Set Super Farad Capacitor 16V 16.6F High Current Ultracapacitor for Board for for Car Automotive Rectifi, 12 x 2 x 6.2cm, Blue

Brand: JAGTRADE

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Capacitance	16.6 Farad
Brand	JAGTRADE
Material	Electronic Component

Our secondary capacitor bank consists of six 10F capacitors in series. Using the formula for equivalent capacitance, we get

- 1/Ceq = 6(1/10)
- 1/Ceq = 0.6
- Ceq = 1/0.6 = 1.6F

1 Pcs 16V1F/2F 16V 1.6F Farad Capacitor Module 2.7V 10F Super Capacitors,with Protection Board

Brand: SaiDian

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Capacitance	1.6 Farad
Brand	SaiDian
Material	Aluminum, Plastic
Shape	Round
Operating Voltage	16 Volts

About this item

PARALLEL

We have 2 capacitor banks in parallel, one of 16F and one of 1.6F. Using the formula for capacitors in parallel, we get

- Ceq = 16 + 1.6
- Ceq = 17.6F

Conceptual

The reason the formula for in series is because current is constant across

Visual

For each capacitor bank, using a capacitance meter, touching the negative and positive ends of the linked capacitors measures 16F for the large bank and 1.6F for the smaller bank.

Once we link the capacitors together in series, using a capacitance meter, touching the negative and positive ends of the linked capacitors measures 17.4F.

Reasoning

Our calculated equivalent capacitance in parallel for the big and small capacitor in our circuit is within 1% error of our measured capacitance in the circuit with a capacitance meter, so our claim for parallel capacitors holds true. Our calculated equivalent capacitance in series for individual capacitors in the bank matches the industry measured capacitance for the entire bank, and our calculated equivalent capacitance matches the capacitance measured experimentally, so our claim for series capacitors holds true.

Energy of Capacitor, Energy Storage Circuit, Field Energy Density

What Was Justified

The energy stored in the capacitor is 1/2 CV^2. This setup can also be used to demonstrate energy storage in a circuit. For the purposes of simplicity, we will only consider the larger supercapacitor bank, which contains 6 100F capacitors in series, charged to 16V. We approximate its volume using cylinders to calculate the field energy density.

Evidence

Computational

Assuming each of the sub-capacitors are identical, it can be expected that the voltage drop across each is the same, or 16 / 6 = 2.67 V. Then, the potential energy in each is 1/2 (100) (2.67)^2 = 356.4 J, so the total is 6 * 356.4 = 2138 J.
Considering the whole capacitor bank as a single capacitor, the total energy is 1/2 (16.6) (16)^2 = 2124 J.

The volume of the supercapacitors can be approximated by 6 cylinders, whose total volume is:
6 * (0.01)^2 * 3.14 * 0.06 = 0.000113 m^3. Thus, the energy density when charged to 16 V can be approximated by energy / volume = 18800 kJ/m^3.

Conceptual

Intuitively the sum of the individual energies should equal the total energy as a whole because of conservation of energy. The setup also qualifies as an energy storage circuit because a 12V power source's lower but constant current is used to store energy into capacitors, which are better suited for the project's needs of high-current power in short bursts.

Visual

Super Capacitor 6pcs/Set Super Farad Capacitor 16V 16.6F High Current Ultracapacitor for Board for for Car Automotive Rectifi, 12 x 2 x 6.2cm, Blue

Brand: JAGTRADE

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Capacitance	16.6 Farad
Brand	JAGTRADE
Material	Electronic Component

The six capacitors are connected in series one after another yielding an advertised capacitance of 16.6F, which we confirmed with a capacitance meter

Reasoning

The computational evidence strongly suggests the validity of the formula for the energy (at least the power of each term) because the total and sum of individual energies match (apart from rounding error).

Inside a supercapacitor, usually most of the volume is filled with thin sheets of metal that act as coaxial cylindrical capacitors. Thus, the volume estimate should at least be at the right order of magnitude for the area that there is actually an electric field in.

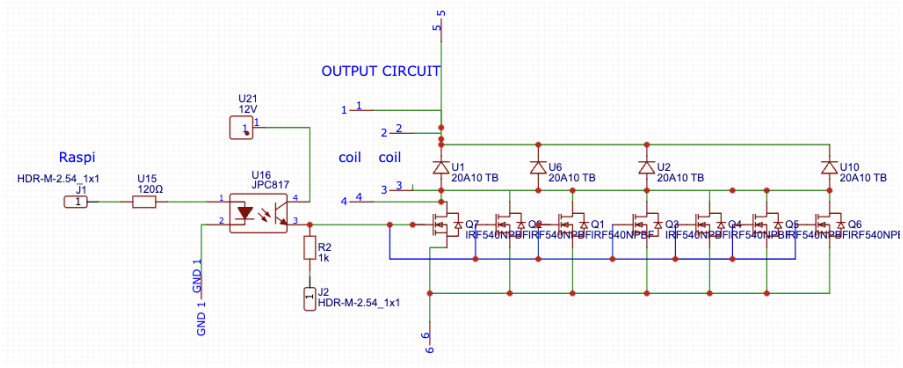
INVESTIGATION 2: Analyzing Our Circuit

Concepts Covered

- Electric Potential KVL & KCL (Voltage drop across the circuit)
- Ohm's Law

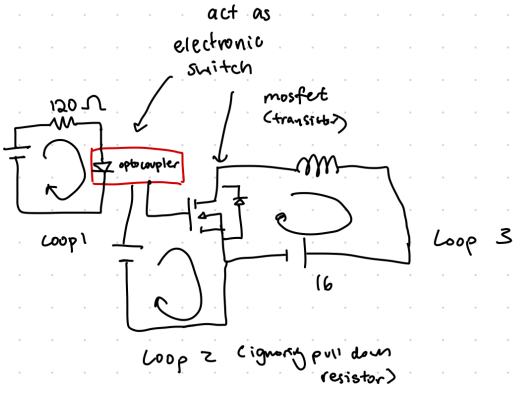
Procedure

A. Parallel Coil Switching



1. Use the circuit board responsible for switching 2 coils in parallel (MOSFETs simplified).

This is a circuit diagram of one of our boards, responsible for switching 2 coils in parallel. We can simplify it to →



(Ignoring the protection diodes and pulldown resistors, combining the parallel MOSFETs)

The Raspberry Pi sends a signal initiating a loop. First, Loop 1 switches on Loop 2, which switches on Loop 3. Focus on loop 3, where current flows through a coil.

2. Measure the voltage drop across the coil during operation.
3. Disconnect the wire before and after the coils.
4. Use an oscilloscope to probe both ends of the disconnected wire for amperage

B. Capacitor Bank KVL Demonstration

1. Use a capacitor bank total voltage.
2. Measure voltage across increasing numbers of capacitors using a multimeter.
3. Record total voltage and individual voltage drops

Data

B. Table of values measured from the multimeter:

Across how many Capacitors	Voltage	Voltage Drop across each Cap
0	0	
1	2.56	2.56
2	5.15	2.59
3	7.76	2.61
4	9.39	1.63
5	11.98	2.59
6	14.57	2.59

Analysis

KVL

What Was Justified

As stated by Kirchhoff's Voltage Law, the voltage gain and drop across a closed loop will sum to zero.

Visual Evidence

To determine the strength of the magnetic field, we needed to find the current across a coil. This is where we used KVL, and we knew that the voltage drop across the 13-16V could be the coil.

Measurement across the parallel coils →

Computational Evidence

Series Capacitors

We simplified loop 3 to being a voltage source powering a coil, and while that is not incorrect, it doesn't show that the 16V is not supplied from a battery, but rather a capacitor bank. As we measure across each capacitor, we observe a voltage drop on the multimeter. If we sum up the voltage drops across each capacitor, we arrive at the voltage of the entire bank.

Vtotal - Vdrop1 - Vdrop2 - Vdrop3 - Vdrop4 - Vdrop5 - Vdrop6 = 0
14.57 - 2.56 - 2.59 - 2.61 - 1.63 - 2.59 - 2.59 = 0

Conceptual Evidence

KVL - As charge flows through a circuit, it is being used up by electrical components. We can observe this as a voltage drop, with a greater resistance of an electrical component causing more voltage drop, as supported by Ohm's Law.

Reasoning

As shown in the computational evidence, by calculating change in the voltage across the complete cycle of loop 3, the total is 0 as KVL dictates. This also confirms that the capacitors are wired in series and not parallel because otherwise each capacitor's voltage difference would be identical and equal to the total to the overall bank voltage.

KCL

What Was Justified

As stated by Kirchhoff's Current Law, the current in will be equal to the current out in a closed circuit.

Visual Evidence

To demonstrate KCL, we disconnected the wire before and after the coils, and put an oscilloscope probe between the 2 disconnected ends. We used an oscilloscope because the short firing time cannot be measured with a multimeter. We measured 42A both in and out.

Conceptual Evidence

KCL - Because charge is conserved, charge cannot accumulate at a node, and just passes through it. This makes the total sum of currents coming into the node equal to the total sum coming out

Computational Evidence

Using the voltage drop from above, we used it to calculate the current across the coil by using Ohm's law.

$V = IR$

$I = V / R = 14.5V \ / 0.32 \text{ ohms}$

$= 45.3A$

This aligns with our measured current: I_in = 43 A, I_out = 43 A. The values are not exactly the same. As the measurements were conducted in different trials, the voltage on the capacitor could have dropped.

INVESTIGATION 3: Analyzing Coil (Inductor) Charging/Discharging

Concepts Covered:

- Back EMF
- Magnetic Energy Stored In Inductor
- Flyback Diodes
- Non ohmic material
- Inductors in Parallel
- Inductor Charging Time Constant
- Inductance from Geometry, Wheelers Formula

Procedure

Part A. Both Coils

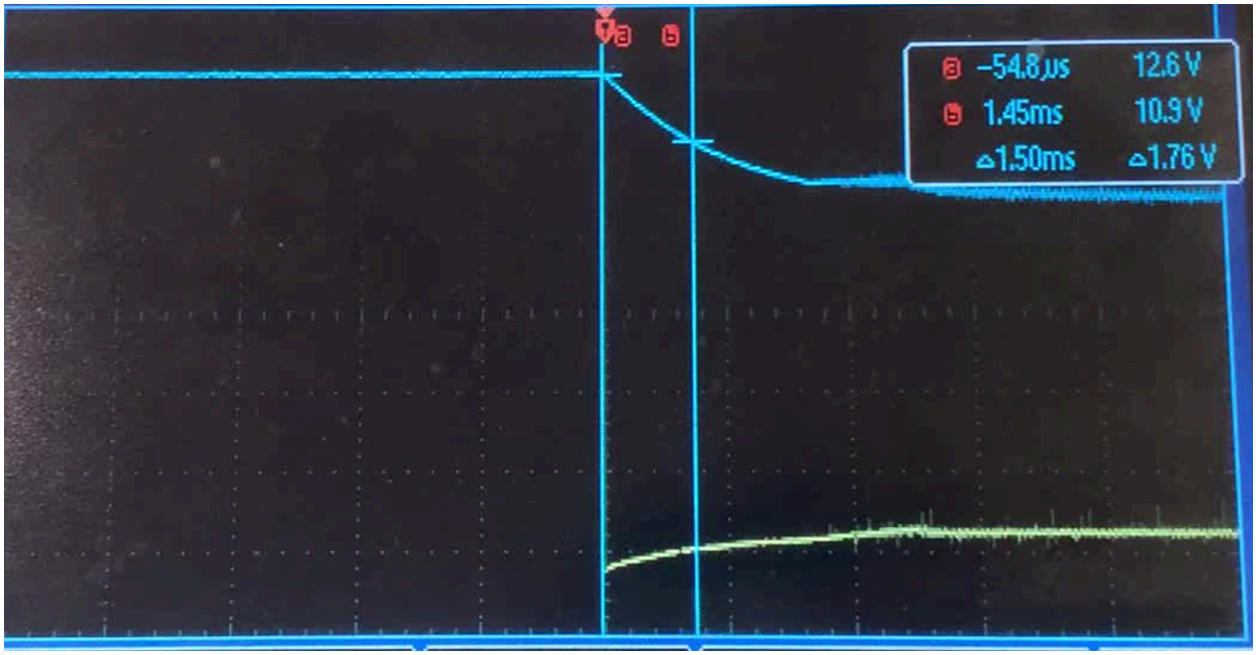
1. Charge the capacitors to 15.6V
2. Connect the positive and negative ends to either end of the inductors or equivalently the opposite ends of the coil
3. Initiate a single sweep with trigger of +5V and windows size of at least 200ms
4. Pulse the coils for 100ms
5. Record the time constant and voltage spikes for the charging/discharging of the coils

Part B. Single Coil

6. Disconnect the left/right coil
7. Charge the capacitors to 15.6V
8. Connect the positive and negative ends to either end of the inductors or equivalently the opposite ends of the coil
9. Initiate a single sweep with trigger of +5V and windows size of at least 200ms
10. Pulse the coils for 100ms
11. Record the time constant and voltage spikes for the charging/discharging of the coils

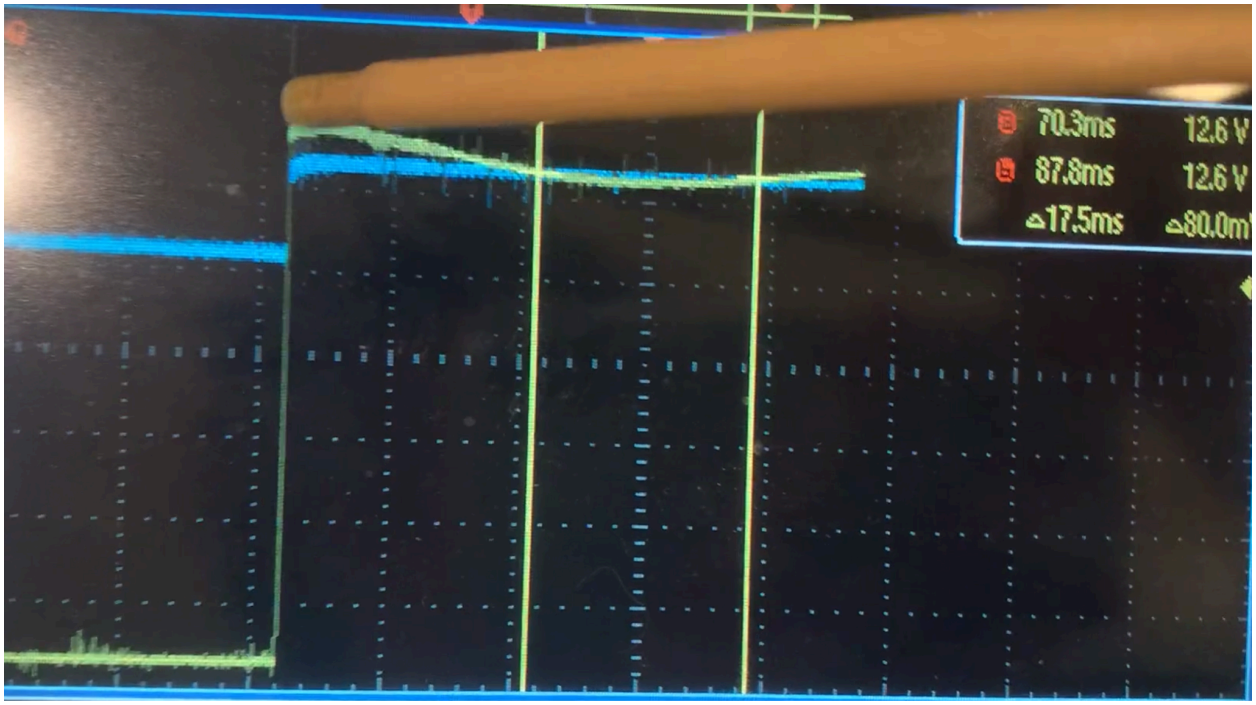
Data

Two Coils LR Inductor Charging V(t) Graph

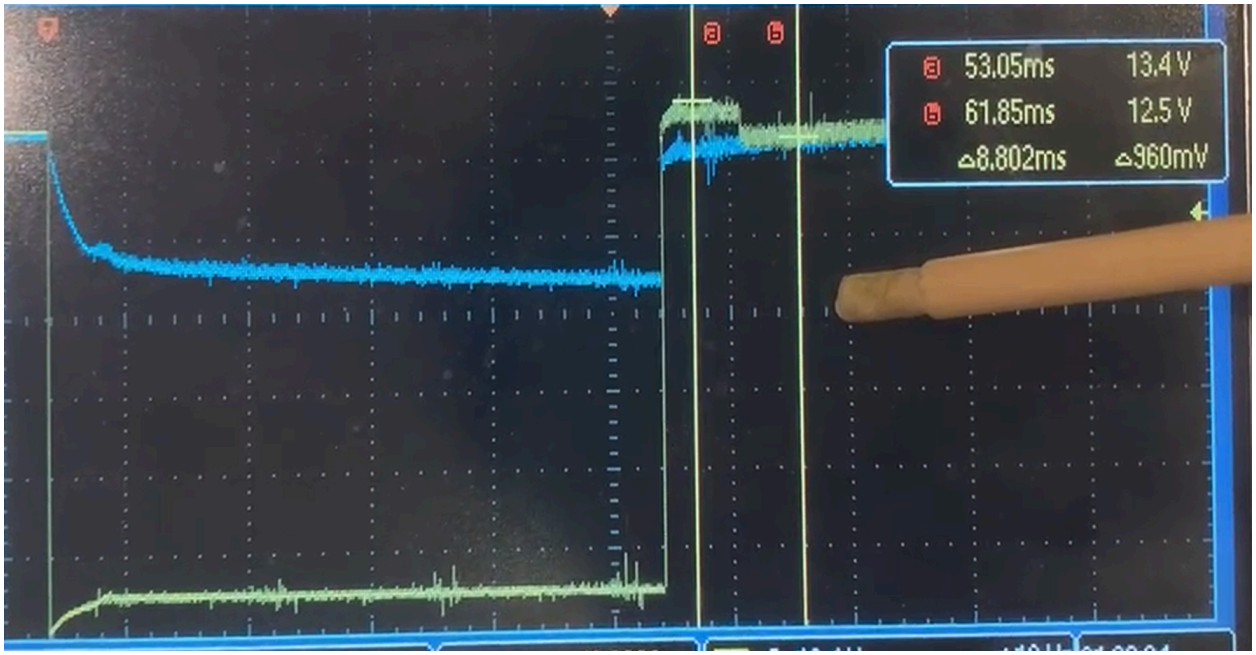


One Coil LR Inductor Charging V(t) Graph

One Coil Back EMF V(t) Graph (840mV drop)



Two Coil Back EMF V(t) Graph(960 mV drop)



Calculation

Computation of inductance from time constants, recorded at 63.2% of the total current.

$$L = \tau \cdot R$$

Case 1: 2 coils
tau = 1.5ms
R = .34ohm

$$L = (1.5 \times 10^{-3}) \cdot 0.34 = 5.1 \times 10^{-4}\,\mathrm{H} = \boxed{510\,\mu\mathrm{H}}$$

Case 2: 1 coil
tau = 1.9ms
R = .5ohm

$$L = (1.9 \times 10^{-3}) \cdot 0.5 = 9.5 \times 10^{-4}\,\mathrm{H} = \boxed{950\,\mu\mathrm{H}}$$

Expected Ratio of L

$$L_{\text{parallel}} = \frac{L_0}{2} \qquad \frac{L_{2\text{ coils}}}{L_{1\text{ coil}}} = \frac{1}{2}$$

- $L_{2\text{ coils}} = 510\,\mu H$
- $L_{1\text{ coil}} = 950\,\mu H$

L_2/L_1 = .537
%Error = **7.4%**

Data/Error Analysis

With our 1 coil inductance of 950 uH and 2 parallel coil inductance of 510 uH we had a very small %error of 7.4% suggesting that our hypothesis that the inductance of two inductors in parallel is accurate.

Non-Ohmic Material, Flyback Diodes

What Was Justified

The flyback diodes in our coilgun are non-ohmic materials, their current-voltage relationship is nonlinear and strongly dependent on the voltage polarity and magnitude

Evidence

Computational

When the diodes have a reverse biased 16 V difference applied across them, they have 0A.
V = IR, R = V/I
V/I = ∞

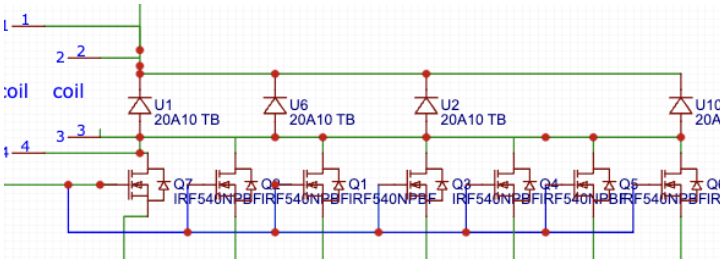
When the diodes have a forward biased Back EMF of 0.9V applied across them, they have an instantaneous I of 11.5A.
V/I = 0.08 Ω

Conceptual

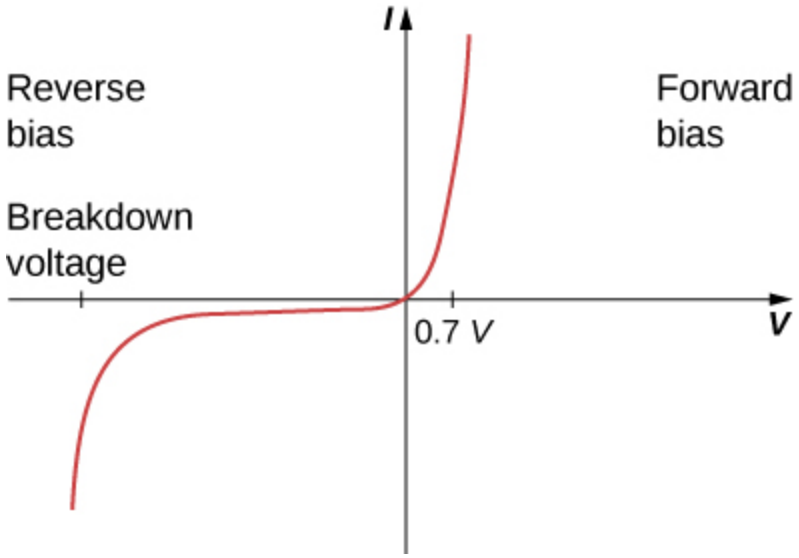
A diode only lets current flow through in one direction, which is a nonlinear I-V relationship, making them non-ohmic

Visual

The flyback diodes (U1, U6, U2, and U10) are connected in parallel across the MOSFET, which allows the current to circulate in the reverse direction to give the inductor's stored energy a path to dissipate



The IV curve on a diode is nonlinear and very little current is let through until it is in the right direction or exceeds the breakdown voltage



Reasoning

Since the resistance of the diodes(V/I) changes with different directions and magnitudes of V, the diode's resistance cannot be represented with one resistance and the diodes are non-ohmic

Back EMF & Magnetic Energy Stored in Inductor

What Was Justified

An inductor will generate a back EMF when current supply is suddenly cut equivalent to the V necessary to generate the current

Evidence

Computational

1. Total current to clamp:
   $I_{total} = V / R = 15.6\ V / 0.34\ \Omega \approx 45.9\ A$

2. Current per diode (4 in parallel):
   $I_D = I_{total} / 4 \approx 45.9\ A / 4 = 11.5\ A$

3. Forward-drop of a 20A10 at ~11.5 A (from datasheet interpolation):
   $V_F \approx 0.85 – 0.90\ V$

4. Peak back-EMF (clamped voltage):
   $V_{clamp} \approx V_F \approx 0.9\ V$

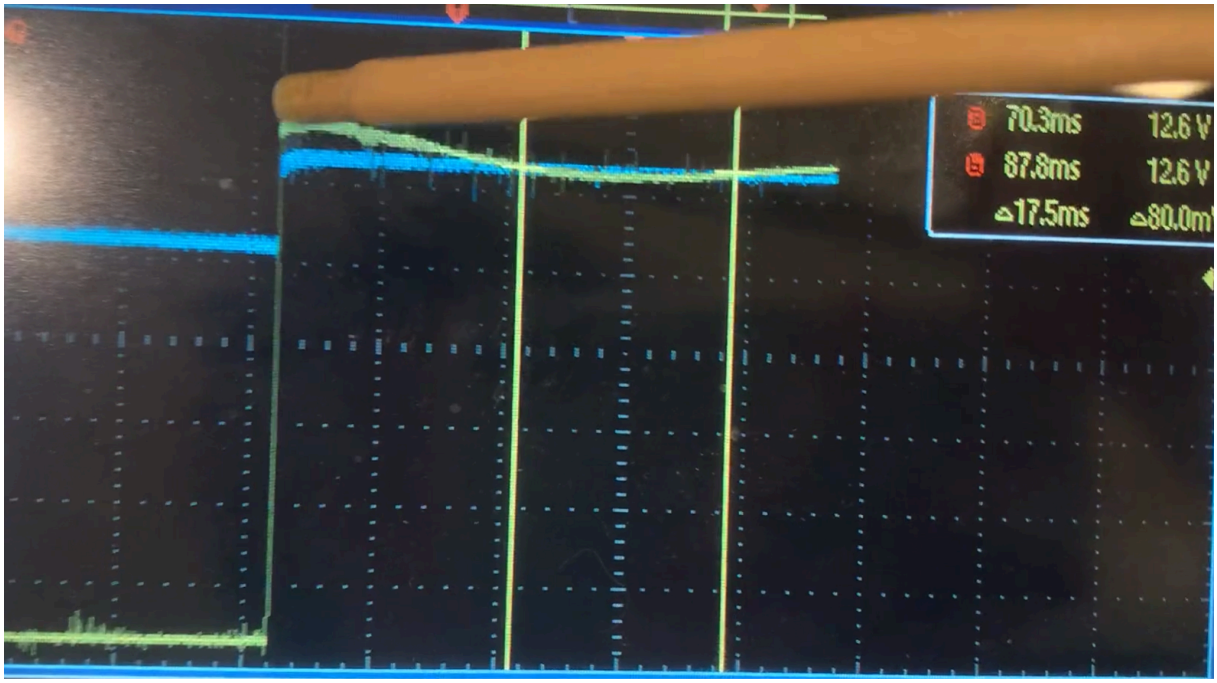
Conceptual

The reason the back EMF exists is because the inductor has stored the current in the form of magnetic field energy and when current stops being supplied, the coils will try to maintain the current equivalent to what it was before. This means that the flyback diodes will instantaneously have the same 45.9A flowing through it, or 45.9A/4 flowing through each diode and which according to the spec sheet should have about 0.9V.

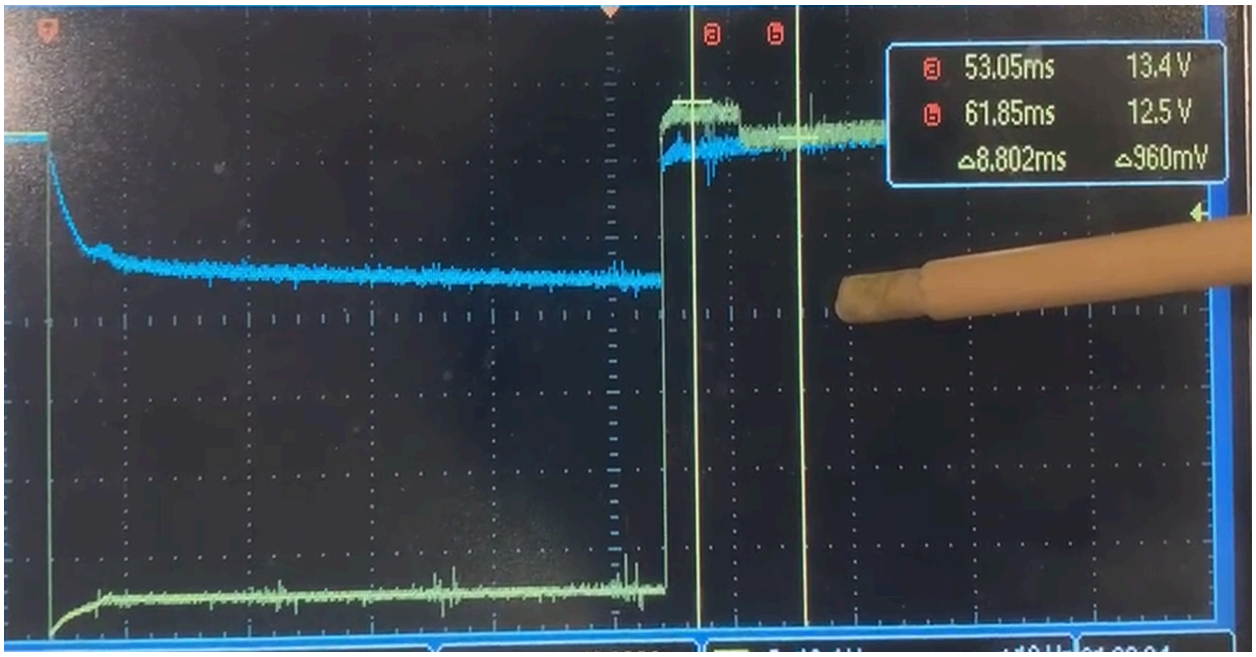
Visual

The 0.9V increase of the negative end of the coil over the positive end of the coil seen in the graph represents the back EMF.

1 coil



2 coil



Reasoning

In order to maintain the current flow that the coils had when current was being supplied. The coils supplied a back EMF through the diodes with a voltage that is just enough to maintain the 45.9A. The 0.9V spike seen in our oscilloscope graph demonstrates this to be the case.

Inductor Time Constant and Magnetic Energy

What Was Justified

The time constant of charging an LR circuit can be represented by L/R, LR circuits charge with an exponential curve.

Evidence

Computational

LR Circuit Charging Derivation

Step 1: Apply Kirchhoff's Voltage Law (KVL)

$V = L * (dI/dt) + R * I$

Step 2: Rearrange into standard linear form

$(dI/dt) + (R/L) * I = V/L$

Step 3: Solve using integrating factor method

Let $\alpha = R/L$

Equation becomes:

$(dI/dt) + \alpha * I = V/L$

Integrating factor $\mu(t) = e^{(\int \alpha\, dt)} = e^{(\alpha t)}$

Multiply both sides by $\mu(t)$:

$e^{(\alpha t)} * (dI/dt) + \alpha * e^{(\alpha t)} * I = (V/L) * e^{(\alpha t)}$

Left-hand side is the derivative of $(e^{(\alpha t)} * I)$:

$d/dt [e^{(\alpha t)} * I] = (V/L) * e^{(\alpha t)}$

Step 4: Integrate both sides

$\int d/dt [e^{(\alpha t)} * I] dt = \int (V/L) * e^{(\alpha t)} dt$

$e^{(\alpha t)} * I = (V/L) * \int e^{(\alpha t)} dt$

$e^{(\alpha t)} * I = (V/L) * (1/\alpha) * e^{(\alpha t)} + C$

$e^{(\alpha t)} * I = (V / (\alpha L)) * e^{(\alpha t)} + C$

Divide both sides by $e^{(\alpha t)}$:

$I(t) = (V / (\alpha L)) + C * e^{(-\alpha t)}$

Recall $\alpha = R/L \rightarrow \alpha L = R$

So:

$I(t) = (V/R) + C * e^{(-t * R/L)}$

Step 5: Apply initial condition $I(0) = 0$

$I(0) = (V/R) + C * e^{(0)} = 0$

$(V/R) + C = 0 \rightarrow C = -V/R$

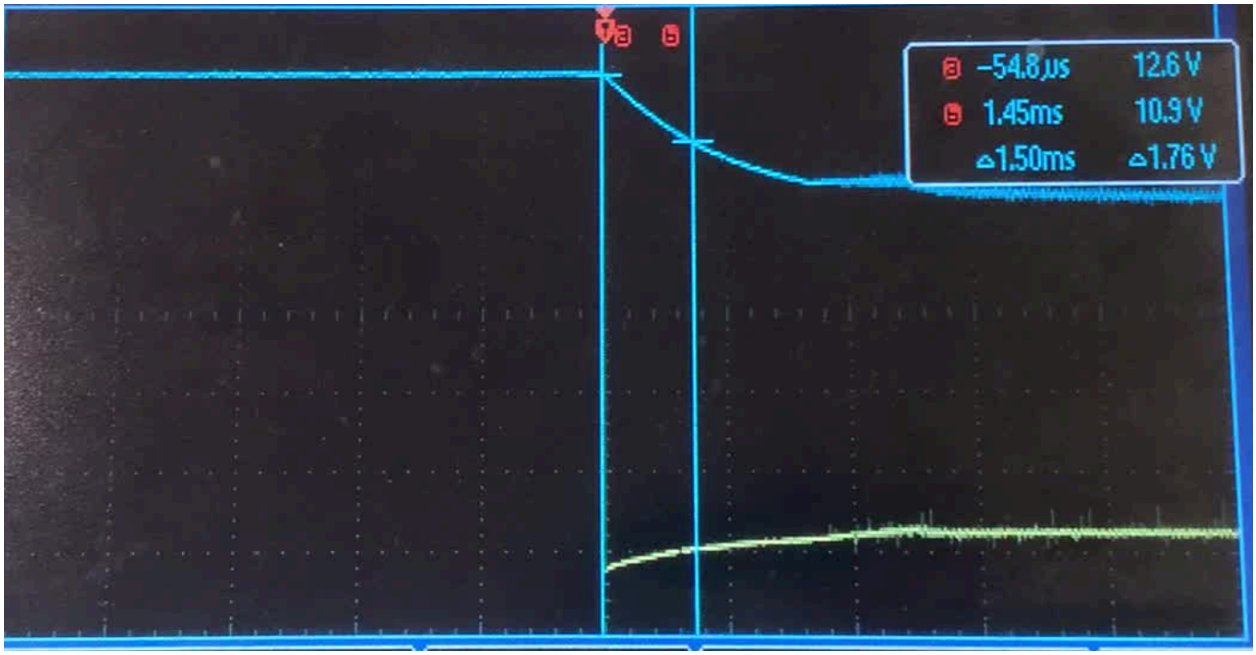
Final solution:

$I(t) = (V/R) * (1 - e^{(-t / \tau)})$ where $\tau = L/R$

Conceptual
It makes sense that the LR circuit charging follows this increasing exponential curve because when the power is first applied the dI/dt is very large, so it will prevent all the current change, then as current increases, the voltage difference decreases and the dI/dt also decreases.

Visual
In this graph the inductor is being charged and we can see that the voltage drop, which is proportional to the I coming out of the capacitor, is increasing with an exponential curve

that matches our graph and a time constant of about 1.5ms.



Reasoning

Our derived computational model shows that the charging of the inductor is exponential with a high rate of charging at the beginning that quickly decays off. This makes sense conceptually as the difference between the instantaneous and steady state current decreases the dI/dt will decrease and the rate of charging of the inductor will decrease. Looking at the oscilloscope graph reveals that this relationship is accurate.

Inductors in Parallel

What Was Justified

The inductance of two inductors in parallel can be expressed as $L_1L_2/(L_1+L_2)$

Evidence

Computational

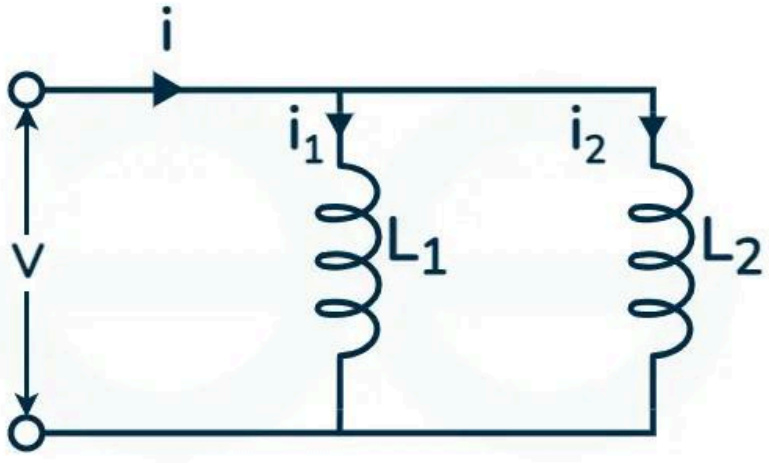
The EMF generated by an inductor in response to a voltage change is -LdI/dt. When the current is split between two inductors in a junction. We can split I' into I_1' and I_2'.
Since V = -LI', plugging in this split gives:
$V = -L_1I_1'$, $V = -L_2I_2'$
$I_1' = -V/L_1$, $I_2' = -V/L_2$
$I' = I_1' + I_2' = -V/L_1 -V/L_2$
$I' = - V/L$
$L = L_1L_2/(L_1+L_2)$
This matches our measured results with 2 coils having 510 uH , while 1 coil had 950uH. Which only has a 7.4% error.

Conceptual

This makes sense because as current is split between the two inductors, the dI/dt is split, so it becomes smaller, creating a smaller EMF and a smaller effective L.

Visual

The current splits into i1 and i2 reducing the effective di/dt



Reasoning

Our derivation calculation shows that our model is accurate and when cross checked with our measured time constants we find a relatively small 7.4% error. Which is very low, especially considering that with the millisecond scale the resolution of measurements are not very high. Additionally this makes sense conceptually because current is split, reducing the effective dI/dt, and reducing the EMF created by the inductor.

Inductance From Geometry: Wheeler's Formula

What Was Justified

The inductance of a multi-layered solenoid can be roughly modeled by Wheeler's Formula:

$$L(\mu H) = \frac{0.8 \cdot r^2 \cdot N^2}{6r + 9\ell + 10d}$$

Evidence

Computational

Coil dimensions:

Inner diameter (ID): 0.25 in
Outer diameter (OD): 2.25 in
Length (ℓ): 0.6 in
Wire gauge: 18 AWG ≈ 0.0403 in diameter

Calculations:

Avg diameter: 1.25in
Avg radius = 0.625 in
Radial depth = 1.0 in
Radial Layers = 1.0/0.0403 = 25 layers
Turns per layer: 0.6/0.0403 = 15 turns

Wheeler's Formula

$$L(\mu H) = \frac{0.8 \cdot r^2 \cdot N^2}{6r + 9\ell + 10d}$$

$$L = \frac{0.8 \cdot (0.625)^2 \cdot (375)^2}{6(0.625) + 9(0.6) + 10(1.0)}$$

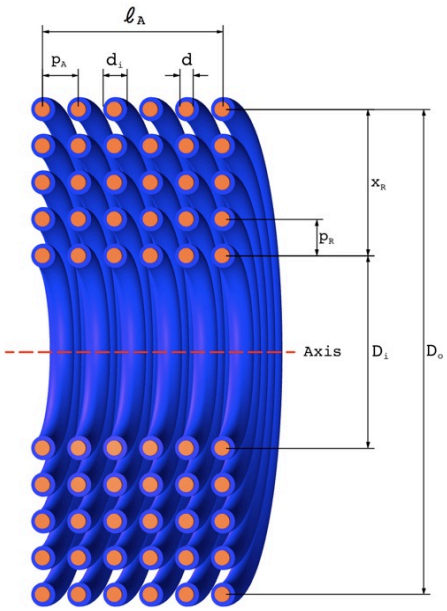
L = **2.29mH**

Error Analysis:

We measured an inductance of 1 coil of 950 uH which has 58.5% error. Some possible sources of error are the internal resistance of the coil as well as inaccuracies in our measurements.

Conceptual
More turns leads to more magnetic field generated by current flowing through the inductor so it makes sense that more turns will increase the inductance

Visual
This diagram shows an approximate geometric model of the inside of a coil. Which we use to estimate the number of turns



Reasoning

Since the computational model had a pretty close estimate to our measured inductance of our coils, with a percent error of 58%, despite all the inaccurate measurements and assumptions. We believe that Wheeler's formula makes computational sense. Conceptually Wheeler's formula makes sense because more turns creates more magnetic field energy and more inductance. Visually, the way the number of turns is approximated also makes sense as shown in the visual model.

INVESTIGATION 4: Analyzing Coil Magnetic Fields

Concepts Covered

- Expected Magnetic Field of a Coil
- Polarization (Iron Cores)
- Moving Charge and Magnetism

Procedure

1. Measure the number of turns of each coil.
2. Measure the length of each coil.
3. Measure the voltage across the capacitor bank.
4. Calculate the coil current using Ohm's Law.
5. Use the magnetic field formula for a solenoid to calculate expected B-field.

Data

Length of Each Coil	0.6 in
Outer Diameter of Coil	2.25 in
Inner Diameter of Coil	0.25 in
Gauge of Wire	18 AWG
Number of turns per layer	15 Turns
Number of layers	25 layers
Total Number of turns	375
Pure Iron Core Permeability	5000
Voltage Across Coils	16V
Resistance Across Coils	0.54 ohm

Expected Magnetic Field of a Coil & Polarization

What Was Justified

Moving charge in a coil (current) generates a magnetic field. The iron cores inside the coils become polarized as the coil is energized, increasing the magnetic force and field strength due to their high permeability.

Evidence

Computational

$$B = \mu \cdot \mu_0 \cdot \frac{N \cdot I}{L}$$

$$B = 5000 \cdot (4\pi \times 10^{-7}) \cdot \frac{375 \cdot 29.63}{0.01524}$$

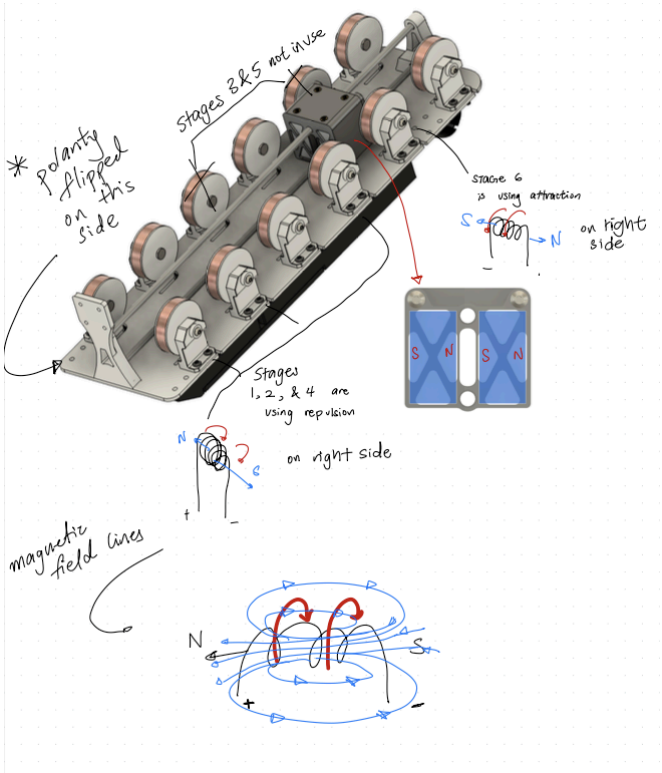
$$\boxed{B \approx 4.58 \text{ T}}$$

This is inside the iron core, not what is exactly interacting with the coil

Visual

The pulse is too small to measure using a phone's magnetic sensor; however, using a compass, we are able to see that the compass needle moves in the direction as expected. We can also see this on the rail. When we energize just one of the coils, with the similar magnetic poles repelling, we observe the rail flexing outwards and the payload (carriage) being tilted.

In order to determine which way the coils should be wired and turned with respect the magnets, we had to determine the magnetic field of the coil.
This can be done with the right hand rule, where you coil your nonthumb fingers in the direction of current and then the thumb would indicate the direction of the magnetic field.



For the first 3 stages (repulsion), we made sure the poles were the same (for ex: N & N, S & S). For the last stage (attraction), we made sure the poles were the opposite (for ex: N & S).

Conceptual

- Moving charge creates a magnetic field because the spinning and orbiting of the nucleus
- As a result, the Iron cores are polarized. We added a ferromagnetic core (iron) to the center of the coil, which becomes magnetized in the presence of a magnetic field, which is called polarization. The iron core (7500) has a higher magnetic permeability than air ($4\pi \times 10^{-7}$ T·m/A), so it can amplify the magnetic field inside the coil.
    - The magnetic dipoles of atoms in the iron align bc of the external magnetic field. It increases the internal field (B), and the core concentrates the magnetic field lines, which focuses the field along the desired path.

Reasoning

Current flowing through a coil generates a magnetic field, and inserting an iron core amplifies this field with its high magnetic permeability. Computationally, using $\mu = 7500$ and $I = 45.3$ A, the magnetic field is 1.07 T. Visually, this is supported by compass deflection and rail flexing when the coil is energized, indicating a strong magnetic force. Conceptually, this happens because the iron core's magnetic dipoles align with the coil's field, reinforcing it and concentrating the field lines, which increases the overall magnetic strength and direction.

INVESTIGATION 5: Analyzing Moving Magnet Induction

Concepts Covered:

- Magnetic Flux (Calculate theoretical flux)
- Expected Inductance
- Compare to actual inductance

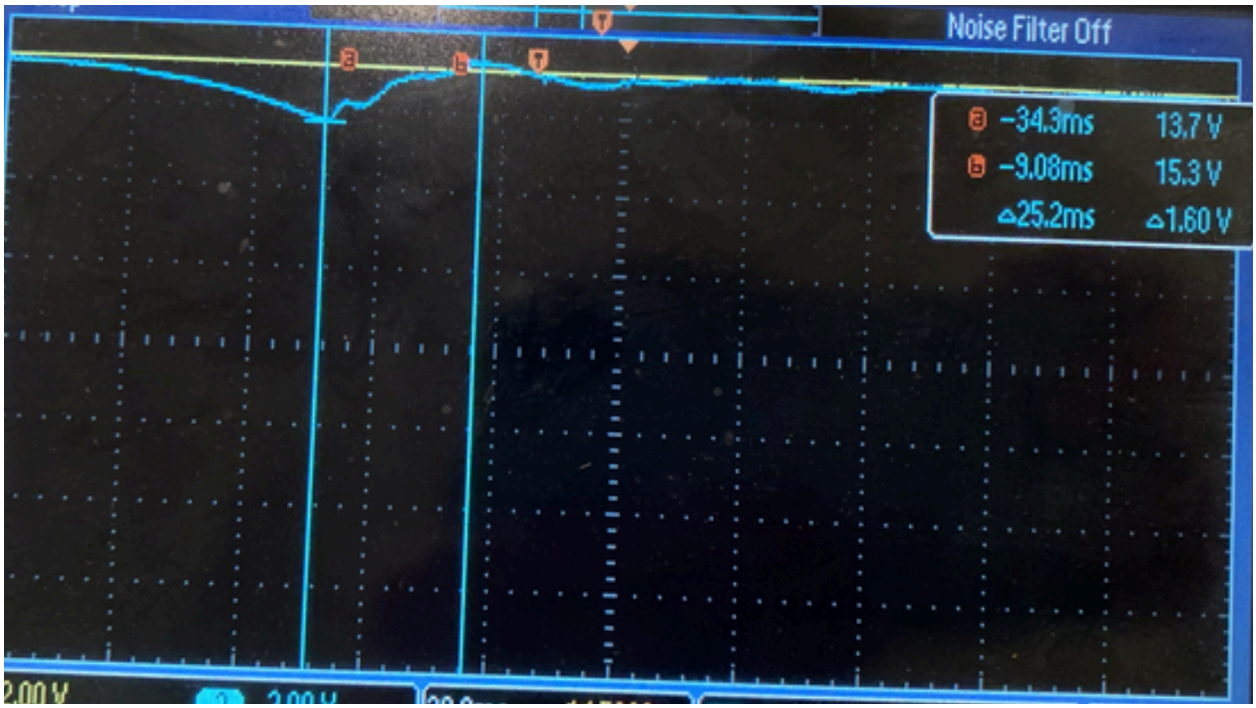
Procedure

Discharge through one coil:

1. Connect the oscilloscope two two ends of a coil
2. Initiate a single sweep with trigger threshold .1V below resting voltage
3. Manually push the magnetic carriage between the carriage
4. Record the Peak induced EMF
5. Record Using Phone magnetometer app the strength of magnetic field of the magnet
6. Record the pulse time from optimized launch sequences to approximate the time it takes for magnet to pass the coil

Data

Moving Magnet Induced EMF V(t) Graph



Peak Induced EMF of 1.6V

Coil Inner Diameter	¼”
Coil Outer Diameter	2.25”
Coil N of turns	375
Strength of magnetic field	1700uT
Avg Area of Coil Loop	$3.17*10^{-3}m^2$
Estimated time it takes magnet to reach center of coil	10ms

Magnetic Flux & Expected Inductance from a Moving Magnet

What Was Justified

Induced EMF can be calculated with Faraday's law:

$$\mathcal{E} = -N \cdot \frac{d\Phi}{dt}$$

Evidence

Computational

Magnetic Flux Calculation

$$\Delta\Phi = B \cdot A = (4.7 \times 10^{-3}) \cdot (3.17 \times 10^{-3}) = 1.49 \times 10^{-5}\ \text{Wb}$$

Rate of change of flux

$$\frac{d\Phi}{dt} = \frac{1.49 \times 10^{-5}}{0.005} = 2.98 \times 10^{-3}\ \text{Wb/s}$$

10ms is divided by 2 in this calculation because of 2 coils rather than 1 coil
Induced EMF

$$\mathcal{E} = -N \cdot \frac{d\Phi}{dt} = -375 \cdot 2.98 \times 10^{-3} = \boxed{-1.12\ \text{V}}$$

Comparison to measured EMF
Measured EMF: 1.6V
%Error = 42.8%

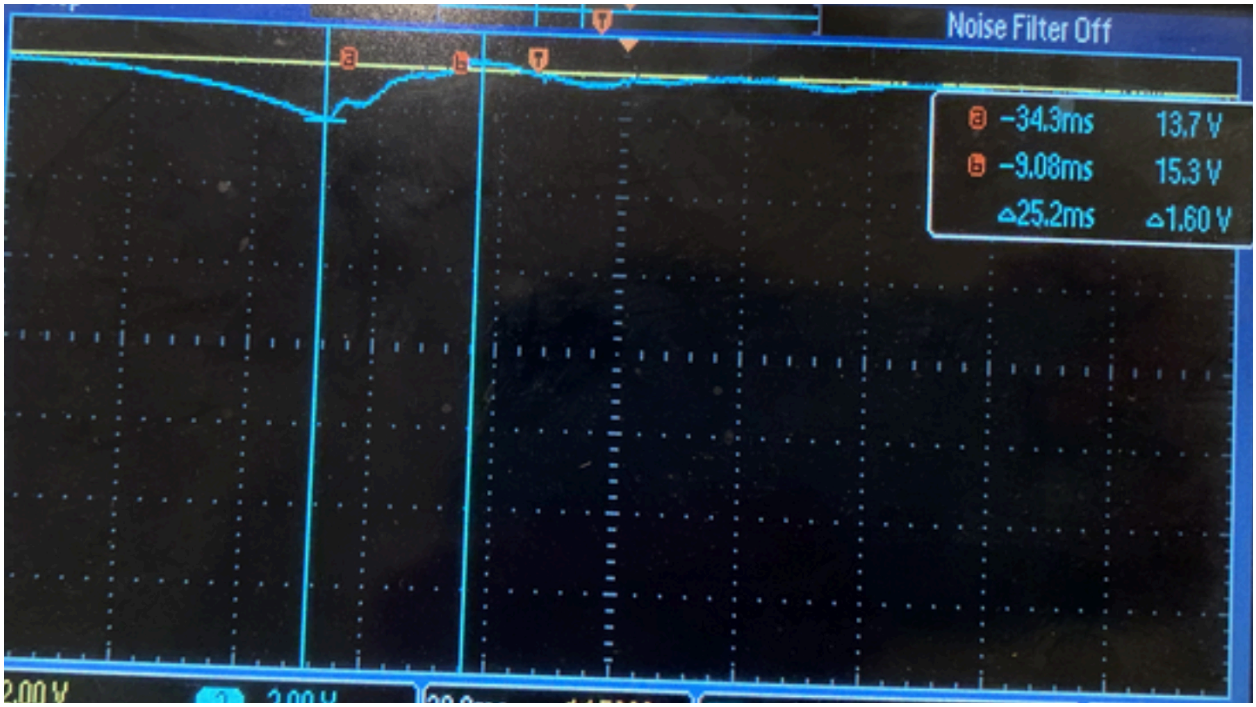
Error Analysis
The possible sources of error are the magnetic field measured from the phone meter being too low, or the estimated velocity of the magnet being too slow

Conceptual

The inclusion of the number of turns in the formula makes sense because more turns means that there will be more loops that have the magnetic flux changed, resulting in a greater rate of total magnetic flux.

Visual

When measuring the voltage across the 2 coils we can see a peak emf of 1.6V



Reasoning

The data and calculations support the idea that a moving magnet through a coil induces an EMF, as predicted by Faraday's Law. Since the EMF is proportional to both the number of turns and the rate of change of magnetic flux, the calculated EMF closely matches the measured peak of 1.6 V, with a reasonable error margin. The magnetic field strength, loop area, and passage time all contribute to the expected voltage, and the inclusion of two coils effectively halves the total time for flux change, increasing the rate. This alignment between theory and observation justifies the conclusion that the changing magnetic flux from a moving magnet is responsible for the induced voltage.

INVESTIGATION 6: Analyzing Our Product Functionality

Concepts Covered:

- Superposition of Two Current Carrying Coils
- Conversion of Electrical to Mechanical energy
- Poynting Vector
- Magnetic Force

Procedure

Part A: Discharge through two coil

1. Charge the capacitor to 12.48V
2. Fire the entire coilgun
3. Record the direction of force
4. Record the distance fired
5. Repeat with voltages from 12.48 - 15.6V with arbitrary increments

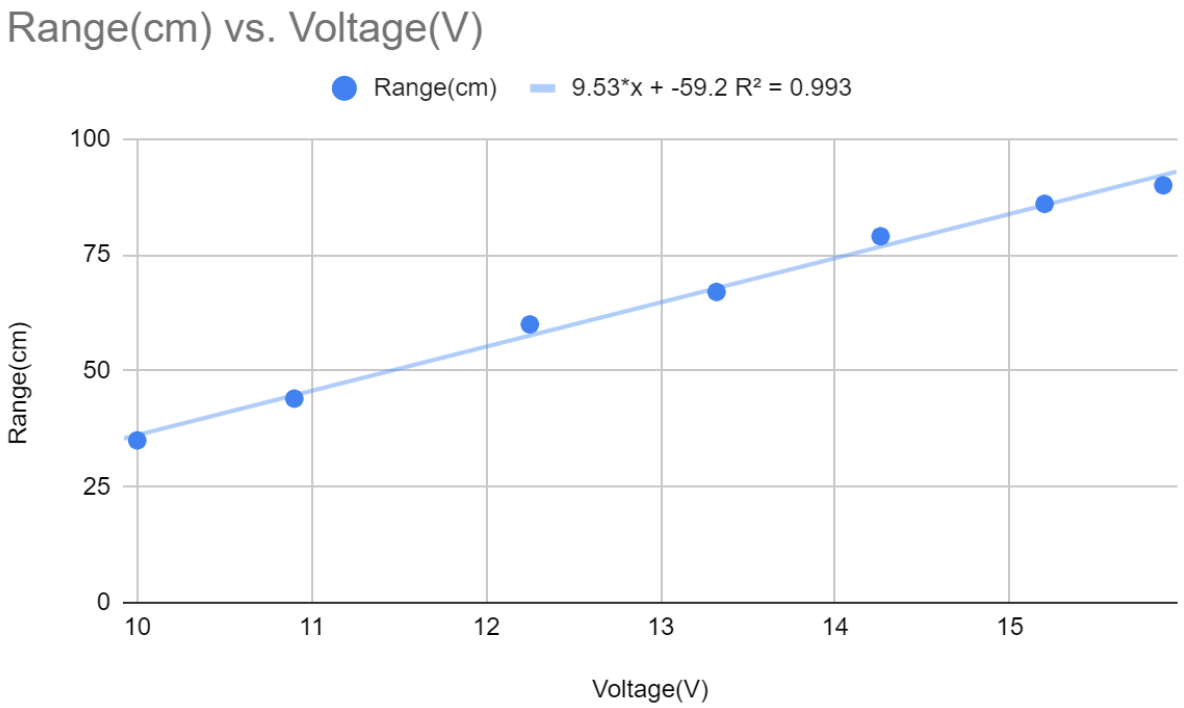
Part B: Discharge through one coil

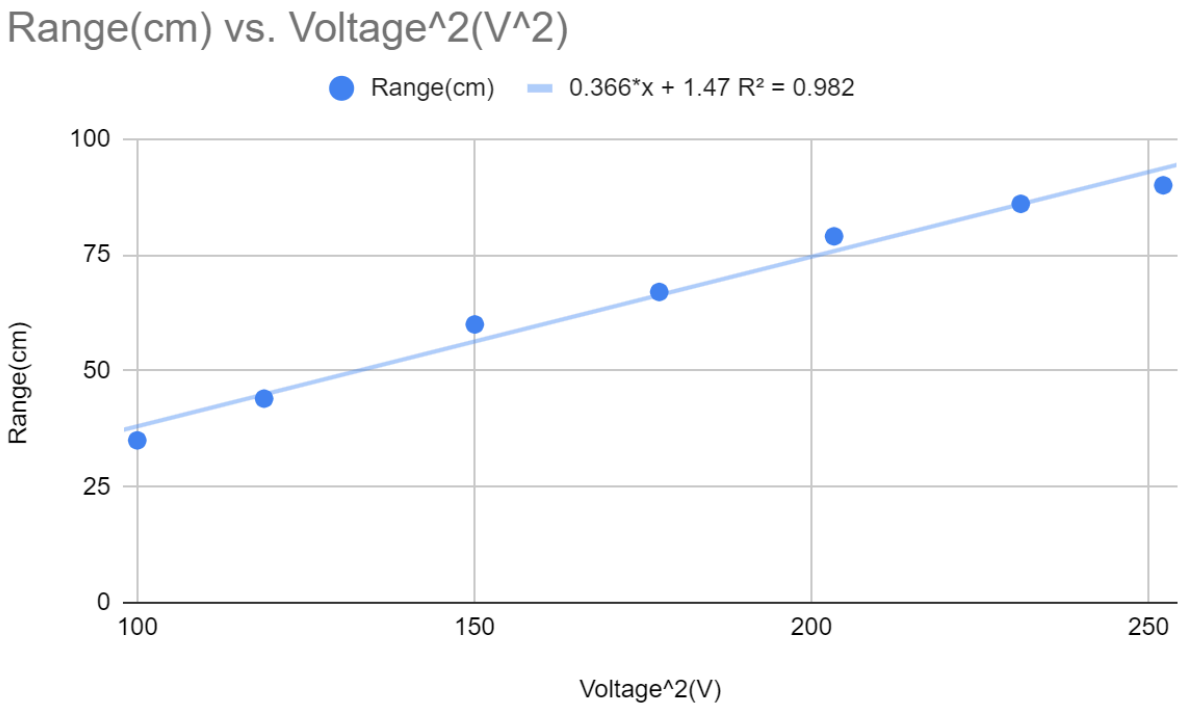
6. Charge the capacitor to 12.48V
7. Disconnect the left/right coil
8. Fire one coil
9. Record the direction of force

Data

Voltage(V)	Range(cm)	Target Pos(cm)	Reached Target	Velocity (m/s)	Kinetic Energy (J) w/ carriage mass = 314g
10	35	38	Yes	1.99	0.62
10.9	45	50	Yes	2.26	0.80
12.25	60	57	Yes	2.61	1.07
13.32	67	60	Yes	2.75	1.18
14.26	79	75	Yes	2.99	1.40
15.20	86	90	Yes	3.12	1.53
15.88	90	100		3.19	1.60

Coil Connected	Direction of force
Left	Right
Right	Left
Both	Forward




Range(cm) vs. Voltage^2(V^2)
Range(cm)
0.366*x + 1.47 R² = 0.982
100
75
50
25
0
100
150
200
250
Range(cm)
Voltage^2(V)

Data & Error Analysis

From the graph we can see a roughly linear relationship between voltage and range. The R^2 value is greater than 0.95, but the y-intercept is negative. This is expected because the range is proportional to the square of the launch velocity. The launch velocity is proportional to the kinetic energy, which is proportional to F x d. In all voltages the d that the force is applied across stays the same, but the forces increases linearly with the magnitude of the electric field. The negative y-intercept can be accounted for by the amount of energy required to get the carriage to move at all and the energy loss that the carriage has when it travels across the length of the rail. Additionally, all 6 trials at different ranges all made it into the basket, giving a **100% success rate**. This is an improvement from the 80% success rate from our last project mainly due to 2 factors: the decreased range means more room for error, while the increased projectile size actually gives it more tolerance to error to still bounce into the basket.

Energy Efficiency Analysis

KE calculation

Velocity is calculated knowing launch angle is 30 deg.

Range = v^2 sin(2(Θ))/g

v = sqrt(Range * g / sin(2(Θ))

KE = 1/2mv^2

= 1/2m(Range * g / sin(2(Θ)) => table of values shown above

Current Calculation

I = (Vi - Vf) / R

I = (12.8 - 8.72) / 0.18 = 22.67A

Electrical Energy = V* I* t =12.48V * 22.67 A* 0.02s * 4 stages = 22.6J

KE_max = 1.60J

To see how much Electrical energy was converted into KE, we can calculate the energy efficiency.

Energy Efficiency = KE_max/Electrical Energy = 1.60J/22.6J= 0.07 => **7% Efficiency**

Sources of Error

This indicates that not all of the electrical energy was converted into mechanical energy. There are losses to the environment, as well as the timing of the reluctance system. Our projectile is not at the strongest point of the magnetic field, as it is outside of the solenoid and far from it. Eddy currents from nearby conductors can generate opposing magnetic fields, which can increase heat losses. A reluctance system requires the coils to turn on and off quickly, which cannot be perfectly precise. Our capacitor also has internal resistance, but beyond energy losses from the electrical components, there could be losses in friction and air resistance as well.

Superposition of 2 Current Carrying Coils and Poynting Vectors

What Was Justified

The magnetic fields and Poynting vectors from two nearby current carrying coils adds up to form one stronger Poynting vector/ magnetic field.

Evidence

Computational

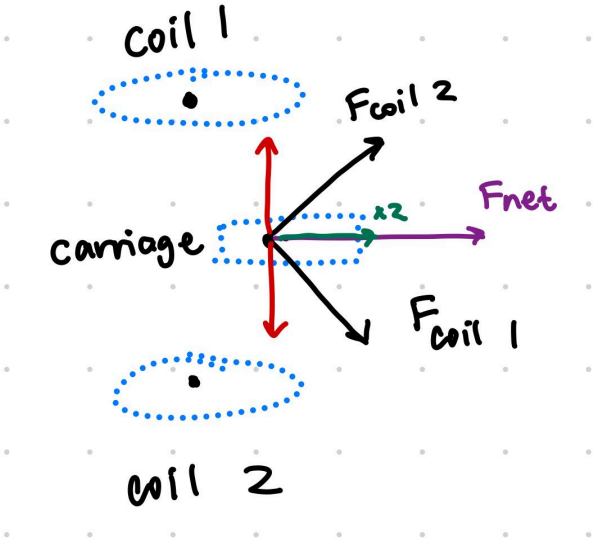
Although only qualitative directional data was collected, we can see that the sum of the left coil and right coil poynting vectors should be approximately the direction of the 2 coil poynting vector.

Coil Connected	Direction of force
Left	Right
Right	Left
Both	Forward

Conceptual

This makes sense because when two forces act on an object what you expect to happen is for the force vectors to add, and this is exactly the same with magnetic force vectors.

Visual



Reasoning

All in all, the 7% energy efficiency (kinetic energy / electrical energy) makes sense, due to being far from the center the magnetic field, inductive inefficiencies, and imperfect timings of each stage. The Lorentz Force equation (F = qv x B) justifies the direction of the magnetic force on the carriage. When both coils are activated, their magnetic fields and the resulting Poynting vectors add up, which produces a forward-directed energy transfer.

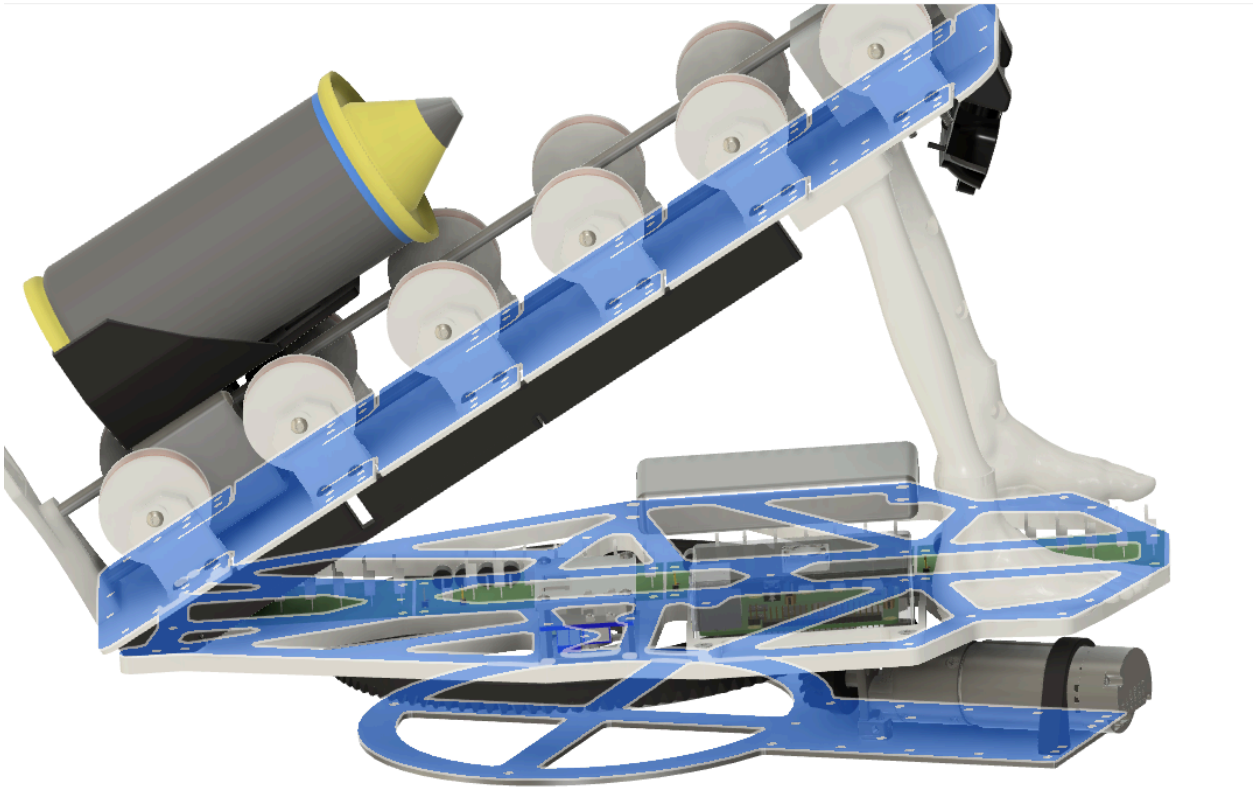
Material Choice

Criteria 9

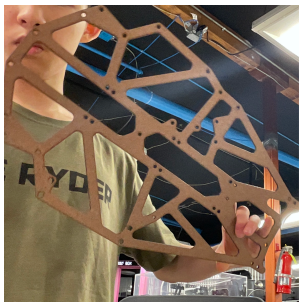
We used the same criteria as last semester to choose our materials, following these considerations: Ease of Manufacturing, Robustness, Availability of tools, Cost

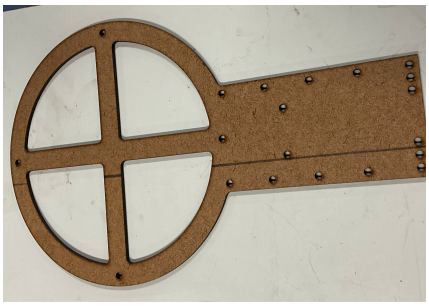
Laser Cutting

This is a manufacturing method not used last semester because of time constraints and not having access to the facilities. However, this semester, we planned our design out earlier, which allowed us to identify two parts that were too large to be manufactured through 3D printing (as highlighted in blue).



While it could be done by joining together several 3D printed parts, it simply isn't practical, so we decided to go to a makerspace to laser-cut it.





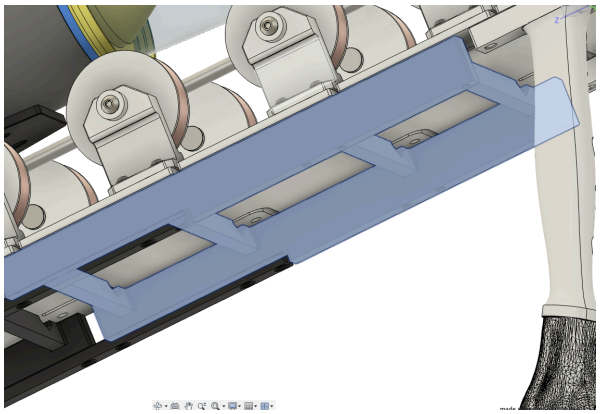
3D printing

While laser-cut wood has an advantage in terms of work area, it has a disadvantage in terms of strength. Therefore, we reinforced it with smaller 3D printed pieces and braces.

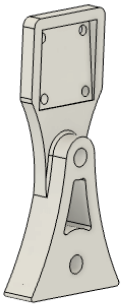
Furthermore, laser-cut wood can only be used to create 2D sheets, while 3D printing can print out nearly any geometry. Therefore, nearly all the custom parts, from the coil holders to the carriage, are created through 3D printing.

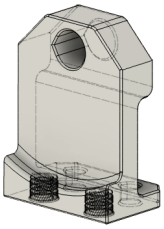
The following are a few examples:

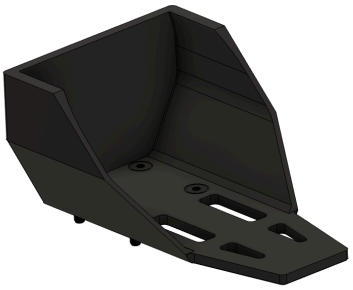
- Wood Reinforcing Braces

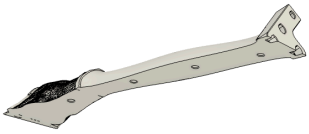


- Complex Geometries





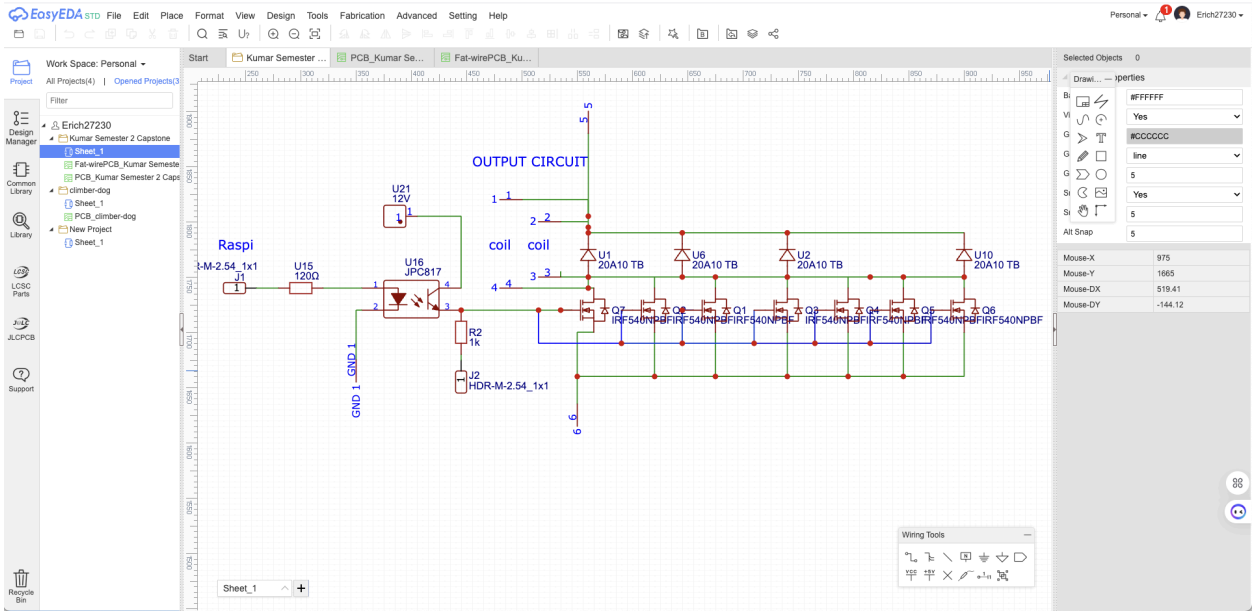




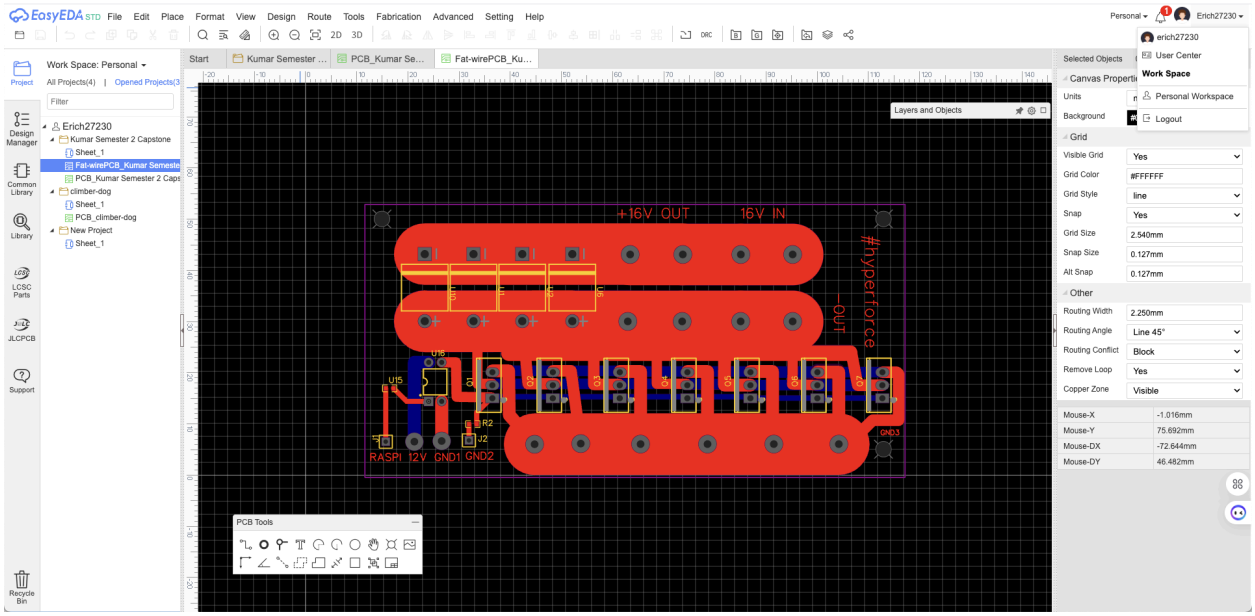
PCBs

We were initially planning 6 stages (12 coils), and we didn't think it was viable to create the circuit by hand, as we would have had to acquire all the individual components and solder them, which would be quite tedious. Therefore, we manufactured it using JLCPCB after designing it using EasyEDA to make the process easier.

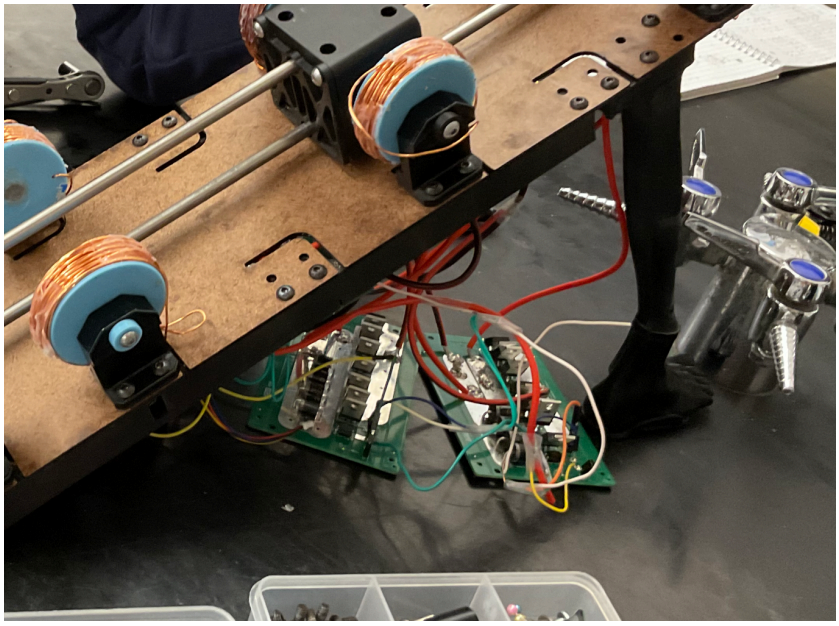
Designed Schematic



PCB Design



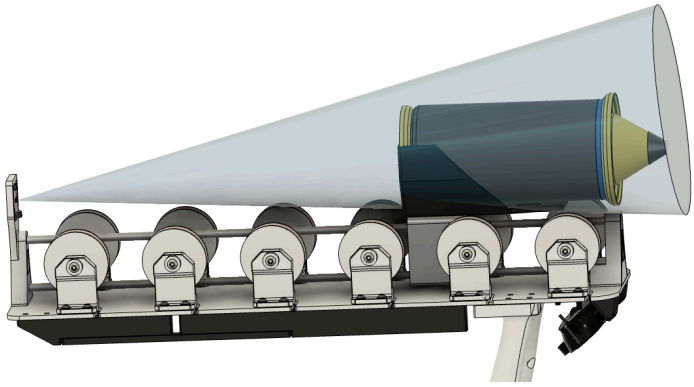
Production PCB



Commercial Off The Shelf Parts

- Electronics Related:
    - Raspberry Pi
        - Raspberry Pi is the brain of the circuit, allowing us to send signals to the PCB to turn on the coils. This would be possible with an Arduino as well; however, we also needed a processor for our sorting computer vision. Raspberry Pi combines those functionalities on one board.
    - 2 Capacitor Banks
        - We underestimated the ESR within the large capacitor, which caused large energy losses. To account for that, we put a capacitor in parallel.
    - Buck-boost converter
        - Allows us convert our input voltage (in this case 12V) to a higher/lower voltage for the capacitors
    - Digital Potentiometer
        - The BuckBoost has a manual 50K W503 potentiometer to control the voltage output.
        - One of the requirements for the launcher is that we can control the launching distance, and to do that, we must vary the voltage on the capacitor. However, there is no way to control the buck-boost

converter's voltage through the W503 potentiometer, so we decided to switch to a digital potentiometer. We desoldered the manual potentiometer and soldered on a 17K ohmic resistor in parallel with a 10K ohm DS332.
        - This allows us to control the capacitor voltages from 12.48 - 15.6V.
    - Adafruit TOF sensor
        - The TOF sensor allows us to measure the distance between the carriage and the sensor.



        - Note that it is tilted up because of its field of view
        - We used this to tune the timings to trigger the coils
- Hardware
    - 16 AWG Magnet Wire
    - 304 Stainless Steel Rods
    - Lazy Susan Bearing
    - Misc hardware: Heatset inserts, screws, nuts
- Parts already on hand
    - Logitech WebCam
    - Motor (Last semester project)

Installation Manual

Criteria 10

Installation steps

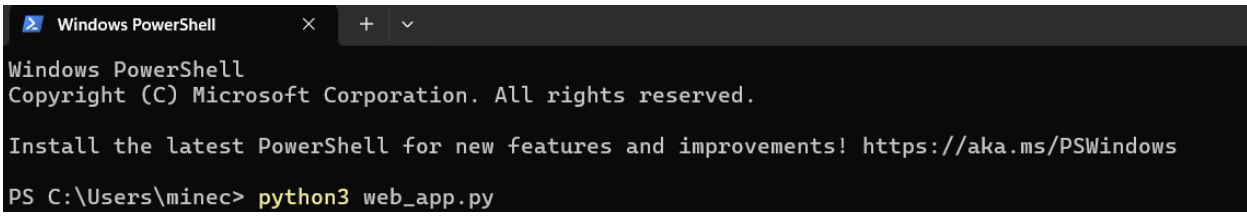
1. Put out the high-friction mat
2. Place the launcher onto the anti friction mat
3. To the right of the launcher, place the power source
4. Plug the power source into an outlet
5. Take out 8 alligator clips
6. Attach alligator clips in the following manner
    a. Buck Boost IN+ -> 12VDC
    b. Buck Boost IN- -> GND
    c. Motor VCC -> 12VDC
    d. Motor GND -> GND
    e. Raspi GND -> GND
    f. White GND wire -> GND
    g. - end of Cap -> 1 ohm resistor
    h. + end of Cap -> Other end of 1 ohm resistor(only when discharging)
7. Turn on the Power source
8. Set up is done!

Usage Manual

Criteria 10

Quick Start Guide

1. Install the necessary software on a laptop.
2. Connect to the raspberry pi.
    a.

    b. Type in the password: raspberry
3. Run the installed app using python and open the web-based control panel.
    a. 

    b. Open up a web browser and type in
        i. http://raspberrypi.local:5000/
4. Discharge the capacitor through a 1 ohm resistor(if already charged too high)
5. Load the screw-on lid capsule with trash and insert it into the carriage
6. Begin the camera homing sequence to detect the location of the target.
7. Launch the capsule when the capacitors are done charging to the right voltage for the target distance.

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Criteria 13

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